

BALASAKTHI SHANMUGARAJA

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PROFILE

Telecommunications Engineering master's student at Politecnico di Milano specializing in **Software Development, mobile networks, RF and wireless communications, network planning, and IoT systems**. Experience with Python-based network optimization, local **AI/LLM** applications, including quantized inference, prompt engineering, API integration, model-response optimization, machine learning, IoT, networking, and wireless technologies.

TECHNICAL SKILLS

Programming: Python, C, MATLAB, SQL, JavaScript.

Tools: Embedded Linux, Git, NumPy, Wireshark, Node-Red, Mosquitto, Pytorch, AWR microwave office, MS office.

Mobile Networks & IoT: TCP/IP, MQTT, ESP-NOW, Network performance Analysis, Network planning and optimization.

EDUCATION

MSc in Telecommunications Engineering

FEB 2025 – Present

[Politecnico di Milano](#)

Track communication network and internet

Average: 24.65/30

B.E. in Electronics and Communication Engineering

Jun 2020 – Jun 2024

[NPR College of Engineering and Technology](#)

CGPA: 7.52/10

Thesis: hybrid vehicle with motion sensors

PROJECTS (Visit [Personal Website](#) for full content)

LLM-Based Automated Question Answering System — *Python, PyTorch, Hugging Face, Qwen2.5-7B*

- Built an automated QA pipeline integrating API-based question retrieval with automated answer generation
- Implemented 4-bit quantization for efficient GPU inference and evaluated performance across question categories

Mobile Radio Network Planning & Optimization — *Python, AMPL, HiGHS, NumPy, Matplotlib*

- Built a MILP-based optimization model for macro/micro base-station placement, balancing coverage, throughput and deployment cost
- Modelled radio propagation using the Okumura-Hata model and benchmarked network performance in Python across multiple deployment scenarios

RF Front-End Design & Performance Analysis — *AWR Microwave Office / VSS*

- Designed and simulated a 4.25 GHz RF receiver front-end (LNA, mixer, filters, local oscillator)
- Evaluated S-parameters, impedance matching, gain, noise figure, IP3, P1dB and EVM, including phase-noise and interference effects

Microwave Network Hardware Failure Classification — *Python, Pandas, Scikit-learn, XGBoost, SMOTEN, CVAE*

- Built a machine-learning pipeline to classify microwave network hardware failures from real and synthetic data
- Applied Random Forest, XGBoost, SMOTEN and CVAE, comparing models on precision, recall and F1-score to handle class imbalance

IoT Systems & Smart Applications — *ESP32, MQTT, Node-RED, LoRa*

- Smart Parking: ESP32 + ultrasonic occupancy sensing over ESP-NOW, with deep-sleep duty-cycling for energy efficiency
- Node-RED/MQTT: built Mosquitto/Node-RED flows for ID/timestamp generation, CSV processing, frame matching and dashboard visualization
- Forklift Monitoring: hybrid indoor/outdoor tracking architecture combining GPS, UWB, IMU, wheel/BLE sensing and LoRa/Wi-Fi/Bluetooth over MQTT

LANGUAGES

Tamil: Native, English: C1 / Advanced, Italian: A2 / Elementary

EXPERIENCE

Python Programming Intern - Phoenix Softech Jul 2023 Completed short-term internship training in Python programming, Embedded Systems Inplant Trainee – Codebind Technologies.